



VAIBHAV TRIPATHI

Research Scholar

Water Resources Development and Management

Indian Institute of Technology Roorkee

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EDUCATION

Doctor of Philosophy (Ph.D.)

Indian Institute of Technology Roorkee, India
Supervisor: Prof. Mohit P. Mohanty

Jan. 2023 to Present

Master of Technology (M.Tech.)

Govind Ballabh Pant University of Agriculture and Technology, India
Supervisor: Prof. Pradeep S. Kashyap

2019 to 2022

Bachelor of Technology (B.Tech.)

Department of Civil Engineering
BML Munjal University, India

2014 to 2018

EXPERIENCE

National Institute of Hydrology (NIH), Roorkee

Project Associate

July, 2022 – Dec 2022

The Glocal University, Saharanpur

Assistant Professor

Jan 2022 – July 2022

M/S G.S. Mehra & Sons

Junior Engineer

July, 2018 – Aug 2019

RESEARCH INTEREST

Hydroclimatic Extremes, Flood Inundation Modelling, Streamflow Forecasting, Remote Sensing and Geospatial Hydrology, Ungauged Basin Hydrology, Physics-aware and Explainable AI for Hydrological Modelling, Monsoon-driven Catchment Hydrology, Climate Change and Water Resources.

TECHNICAL SKILLS

- Hydrologic/Hydrodynamic software: HECRAS, HEC-HMS, SWAT+ and LISFLOOD-FP
- GIS packages: GEE, ArcGIS, Grass GIS and QGIS
- Programming Languages: Python, MATLAB
- Statistical: Machine learning, Deep learning, SPSS

JOURNALS (Published)

1. **Tripathi, V.**, AghaKouchak, A., Karmakar, S., Stein, L., & Mohanty, M. P. (2026). Interpretable deep learning reveals influence and interactions of hydroclimatic drivers on floods across India. *Earth's Future*, 14(7), e2026EF008397. <https://doi.org/10.1029/2026EF008397>
2. Deopa, R., **Tripathi, V.**, Nandana, P. S., & Mohanty, M. P. (2026). Coupled hydrological and public health risks from urban flooding: integrated remote sensing, machine learning, and hydrodynamic–ecological modelling. *Journal of Hydrology*, 135999. <https://doi.org/10.1016/j.jhydrol.2026.135999>
3. Rachit, **Tripathi, V.**, Mohanty, M. P., Pandey, A., & Gupta, A.K. (2026). Rapid Flood Susceptibility Mapping in the Indian Himalayan Region using CNN-U-Net Segmentation: Insights from the 2025 Monsoon Events. *Natural Hazards*, 122(10), 424. <https://doi.org/10.1007/s11069-026-08175-w>
4. Sharma, P.,*, **Tripathi, V.**,*, & Mohanty, M. P. (2026). Improving Daily Streamflow Predictions over Large Watersheds: Introducing a Novel Enhanced Long Short-Term Memory (En-LSTM) model. *Water Resources Management*, 40(2), 92. <https://doi.org/10.1007/s11269-025-04475-1>
5. Deopa, R., Singh, H., **Tripathi, V.**, Tyagi, M., & Mohanty, M. P. (2025). Cascading impacts of 2025 June-July multi-hazard episodes in the Western Himalayas: Evidence through remote sensing observations and Google Earth Engine. *Natural Hazards*, 121(19), 23417-23422. <https://doi.org/10.1007/s11069-025-07699-x>
6. Alam, M. G., **Tripathi, V.**, Bhatt, C. M., & Mohanty, M. P. (2025). A Novel Framework Embedding Bayesian-optimized Ensemble Machine Learning and Explainable Artificial Intelligence (XAI) to Improve Flood Prediction in Complex Watersheds. *Environmental and Sustainability Indicators*, 27, 100760. <https://doi.org/10.1016/j.indic.2025.100760>
7. **Tripathi, V.**, & Mohanty, M. P. (2025). Can geomorphic flood descriptors coupled with machine learning models enhance in quantifying flood risks over data-scarce catchments? Development of a hybrid framework for Ganga basin (India). *Environmental Science and Pollution Research*, 32(59), 31475-31494. <https://doi.org/10.1007/s11356-024-33507-3>
8. **Tripathi, V.**, Mohanty, M. P., & Singh, H. (2024). Fidelity of machine learning models in capturing flood inundation through geomorphic descriptors over Ganga sub-basin, India. *European Water*, 87-88, 1-10. https://www.ewra.net/ew/pdf/EW_2024_87-88_01.pdf

JOURNALS (Under review)

1. **Tripathi, V.**, & Mohanty, M.P. (20XX). Regime-dependent performance of baseflow separation methods across monsoon-dominated catchments in India. (revision submitted, *Environmental Research Letters*, IOP).
2. **Tripathi, V.**, & Mohanty, M.P. (20XX). Nationwide Daily Baseflow Estimation in India (1980–2020) Using Temporal Fusion Transformers and Process-Informed Clustering. (To be submitted to *Scientific Data*, Nature).
3. **Tripathi, V.**, & Mohanty, M.P. (20XX). Do Cluster-Specific Deep Learning Models Outperform Global Architectures in Multi-Horizon Streamflow Forecasting? Insights from Temporal Fusion Transformers Across Indian Catchments. (To be submitted to *Journal of Hydrology*, Elsevier).

CONFERENCES

1. **Tripathi, V.**, Manfreda, S., Samela, C., Navarro, J.S., & Mohanty, M.P. (2026). Regime-Dependent Evaluation of the Geomorphic Flood Index Using ERA5-Driven Hydrodynamic Simulations and Satellite Inundation over a Large Monsoon Basin, **IEEE-IGARSS 2026**, Washington, D.C., USA, 09 August – 14 August 2026. [*received a student travel grant from GRSS and ANRF, India*]
2. **Tripathi, V.**, Singh, H., & Mohanty, M.P. (2026). Ungauged Tributary Discharge Reconstruction for Monsoon Flood Inundation Modelling: A Geomorphic-Recession Approach Applied to Peninsular India, **EGU General Assembly 2026**, Vienna, Austria, 03 May – 08 May 2026, EGU26-815.
3. **Tripathi, V.**, Shaw, A., & Mohanty, M.P. (2025). Nationwide Daily Baseflow Estimation in India (1980–2020) Using Temporal Fusion Transformers & Process-Informed Clustering, **AGU Annual Meeting**, December 15-19, 2025, New Orleans, USA.
4. **Tripathi, V.**, Karmakar, S., Aghakouchak, A., Mohanty, M.P. & Stein, L. (2025). Floods at the Crossroads of Climate & Human Impact: A Process-Based, Explainable AI Framework for Indian Catchments, **AGU Annual Meeting**, December 15-19, 2025, New Orleans, USA.
5. **Tripathi, V.** & Mohanty, M.P. (2025). Interpretable AI for Flood Process Attribution & Prediction in Indian Catchments under Climate & Anthropogenic Influence, **5th International Vietnam Conference on Earth & Environmental Sciences 2025** (iVCEES-2025). [*received a student travel grant to present*]
6. **Tripathi, V.**, Deopa, R., & Mohanty, M. P. (2025). Unraveling Hydrometeorological Drivers of Floods: A Data-Driven Analysis Across India's River Catchments, **EGU General Assembly 2025**, Vienna, Austria, 27 Apr–2 May 2025, EGU25-15096.
7. **Tripathi, V.** & Mohanty, M. P. (2024). A Hybrid Framework of integrating Geomorphic Flood Descriptors & Machine Learning Models to account Flood Risk over an ungauged basin, **HIWeather Final Conference 2024**, 09 September 2024 to 13 September 2024. Munich, Germany [*received a student travel grant from the World Meteorological Organisation, WMO*]
8. Mohanty, M., & **Tripathi, V.** (2024). Can Catchment attributes coupled with an Ensemble of Machine Learning improve Flood Hazard mapping over large data-scarce catchments? **EGU General Assembly 2024**, Vienna, Austria, 14 April – 19 April 2024, EGU24-9063.
9. **Tripathi, V.** & Mohanty, M. P. (2023). Toward Quantifying Flood Hazards over Large Flood-prone River Basins through an ensemble of Geomorphic Flood Descriptors & Machine Learning models, **AGU Annual Meeting**, December 11-15, 2023, USA.
10. **Tripathi, V.**, & Mohanty, M.P. (2023). Employing CORDEX-RCMs for augmenting the performance of Geomorphic Flood Descriptors over large flood-prone & ungauged watersheds. **WCRP Open Science Conference 2023**, October 23-27, 2023, Rwanda.
11. **Tripathi, V.**, Singh, H., Mohanty, M.P. (2023). Fidelity of Machine Learning models in capturing flood inundation through geomorphic descriptors over Ganga sub-basin, India. **12th World Congress European Water Resources Association** (EWRA 2023), June-July 27- 01, 2023, Thessaloniki, Greece.

ACADEMIC HONOURS

- Received the **best oral presentation award** at the Research Scholars Day in the Department of Water Resources Development and Management, IIT Roorkee
- Delivered a **lecture cum hands-on session** on the development of machine-learning models for flood hazard mapping at RASTER 2023 at IIT Roorkee, India

- Working as a **Teaching Assistant (TA)** for River Engineering (RE) and Drinking Water Sanitation (DWS) courses at the postgraduate level at IIT Roorkee.
- Delivered a **lecture cum hands-on session** on the Applications of Machine Learning approaches at the 10-Day Online FDP at NIT Patna on the title “Transforming Water Resources Management, Traffic Systems, and E-Mobility with Quantum Computing and AI”.
- The **Ministry of Human Resource Development fellowship** for pursuing a PhD at IIT Roorkee
- Qualified **Graduate Aptitude Test in Engineering (GATE)** in 2019, 2021 and awarded a scholarship for master's (M.Tech)

TECHNICAL PROJECTS

- Completed a project titled “**Rejuvenation of three rivers flowing through the Bijnor district**” associated with the **National Institute of Hydrology, Roorkee**.

AWARDS

- Received a travel grant to attend and present at the **S2S2D conference** at the **University of Reading**, Reading, United Kingdom, held from September 7-11, 2026
- Received a travel grant to attend and present at the **IEEE International Geoscience and Remote Sensing Symposium** in Washington, D.C., USA, held from August 9-14, 2026
- Received a travel grant from the Dean of Resources and Alumni Affairs, IIT Roorkee, to present at the **European Geophysical Union Assembly 2026** in Vienna, Austria, held from May 3-8, 2026
- Awarded a travel grant from the **World Climate Research Programme (WCRP)** to attend and present a research article at the **5th International Vietnam Conference on Earth & Environmental Sciences 2025**, held in Quy Nhon, Vietnam, from November 24-28, 2025
- Awarded a travel grant from the **World Meteorological Organisation (WMO)** to attend and present a research article at the “**HIWeather Final Conference**” held in Munich, Germany, from September 09-13, 2024
- Awarded a travel grant to attend a one-week basic course on “**Urban Flood Modelling in a Changing Climate**” from June 06-12, 2024, held at BITS Pilani-Hyderabad Campus
- Awarded a travel grant to attend a workshop on “**Application of Machine Learning in Hydrology (Hydro-ML)**, sponsored by **DST**, from March 11-16, 2024, held at IISER Bhopal
- Awarded a travel grant for presenting a poster at the **Fourth Indian Symposium on Machine Learning (IndoML 2023)** from December 21-23, 2023, held at IIT Bombay

MEMBERSHIPS

- International Association of Hydrological Sciences (IAHS)
- Indian Water Resources Society (IWRS)
- Geoscience and Remote Sensing Society (GRSS)
- Institute of Electrical and Electronics Engineers (IEEE)
- American Geophysical Union (AGU)
- European Geophysical Union (EGU)

LEADERSHIP

- Student **Team Coordinator** in the Roorkee Water Conclave, held at IIT Roorkee from 22nd to 25th February 2026
- Elected as a **Hostel Secretary** from 2024-25 at Hostel Affairs, Indian Institute of Technology Roorkee
- Elected as a **Cultural Secretary** from Jan 2024- Aug 2024 at Vigyan Bhawan, Indian Institute of Technology Roorkee
- Appointed as a **Lead Coordinator** for a training program on “Water, Built Environment & Emergency Preparedness” held at IIT Roorkee from 16th to 19th, April 2024
- Worked as an **Organising Head** in the International Conference on Future of Water Resources, held at IIT Roorkee from 18th to 20th, January 2024
- Appointed as an **Organising Head** in a training program on Recent Advances in Satellite-based Technologies for monitoring Hydro-climatological ExtRemes (RASTER 2023) held at IIT Roorkee from 11th to 15th September 2023
- Selected as a **College Ambassador** for two years at BML Munjal University

REFERENCES

- Prof. Amir AghaKouchak, Director, Center for Hydrometeorology & Remote Sensing (CHRS), University of California, Irvine, USA
- Prof. Subhankar Karmakar, Environmental Science and Engineering Department (ESED), Indian Institute of Technology Bombay, India
- Prof. Mohit Prakash Mohanty, Department of Water Resources Development and Management, Indian Institute of Technology Roorkee, India

I, the undersigned, certify that to the best of my knowledge and belief, this biodata correctly describes myself, my qualifications, and my experience.



Vaibhav Tripathi



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Google Scholar